

Machine Learning

Ensemble Methods

FAST

DISCOVERING
THE FUTURE

Topics of previous lectures

- ✓ Ingredients of Machine Learning
- ✓ Classification Basics
- ✓ Basic Linear Classifier
- ✓ K-Nearest Neighbours Classifier
- ✓ Naive Bayes Classifier
- ✓ Decision Trees
- ✓ Support Vector Machines (SVM)

Topics of today's lecture

- Bagging
- Random Forest
- Weighted Voting
- Stacking

Motivation for Ensembles

In 1907, 787 villagers tried to guess the weight of **ox**



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In 1907, 787 villagers tried to guess the weight of **ox**



None of them guessed it correctly, but the **average** guess (**542.9** kg) was very close to actual weight of ox (**543.4** kg)

What are Ensemble Methods?

Definition 1

Combinations of different models are called ensembles.

What are Ensemble Methods?

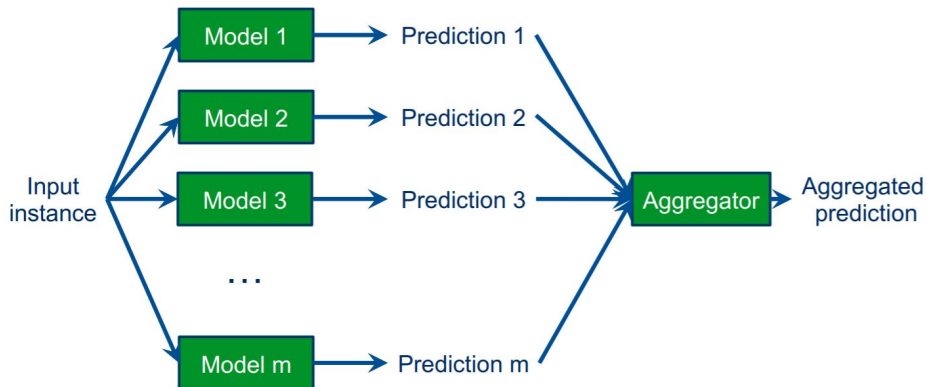
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Definition 2

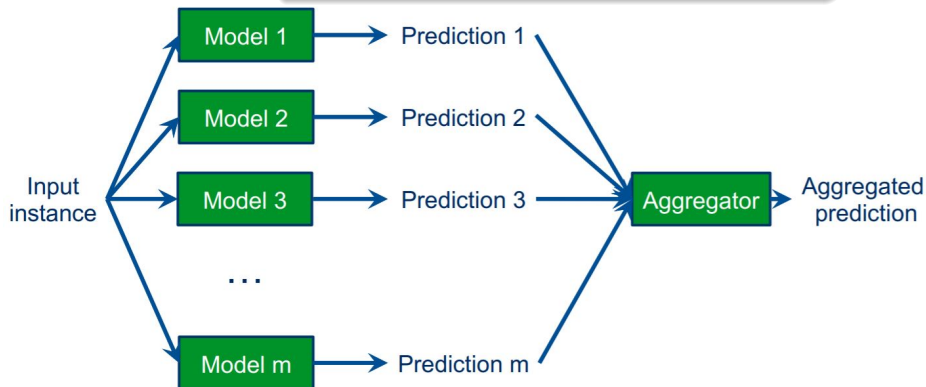
Ensemble method is the one that makes predictions by aggregating predictions from multiple models (ensemble).

Aggregation of predictions



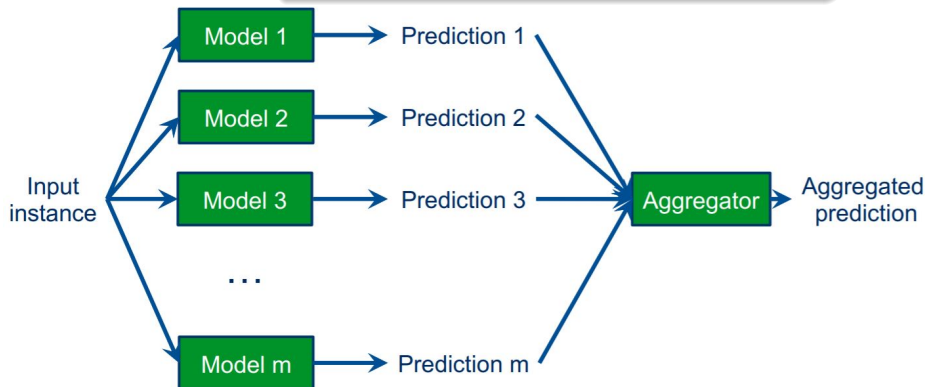
Aggregation of predictions

- How to get multiple models?

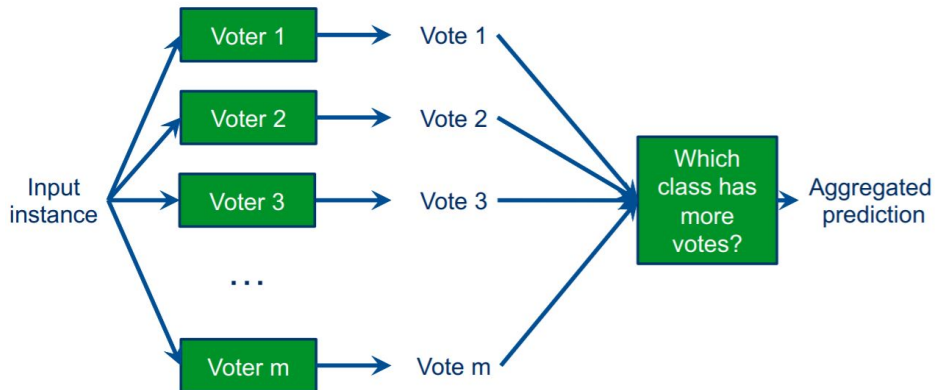


Aggregation of predictions

- How to get multiple models?
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Voting in binary classification



Theory for voting (binary task)

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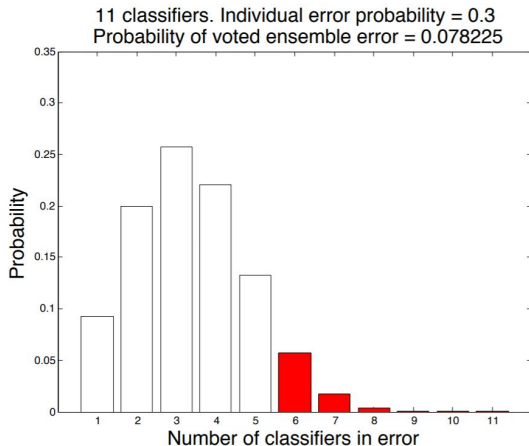
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"Essay on the Application of Analysis to the Probability of Majority Decisions"
- Suppose there are M voters, each having independent errors, and the individual error probability is ϵ
- Majority vote is wrong with probability:

$$\mathbb{P}(\text{majority vote error}) = \sum_{k \geq \lceil \frac{M+1}{2} \rceil} C_M^k \epsilon^k (1 - \epsilon)^{M-k}$$

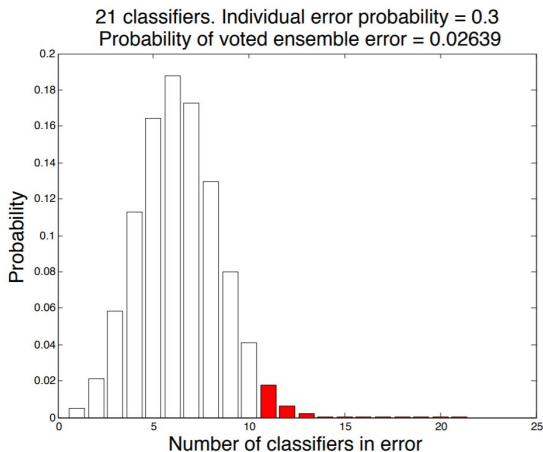
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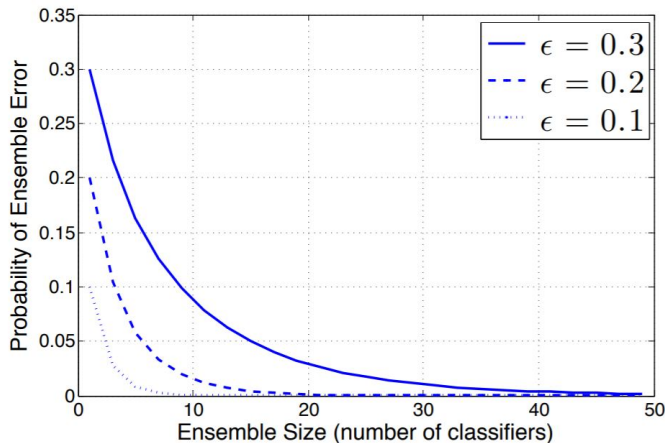
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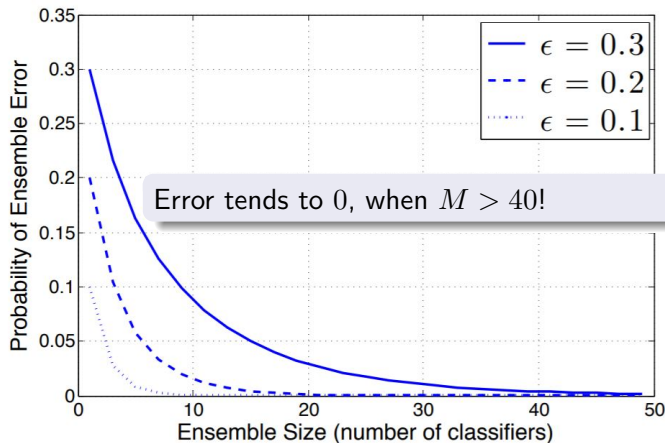
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Why does this theory not work in practice?

Errors of voters are not independent!

Key challenge of ensemble learning

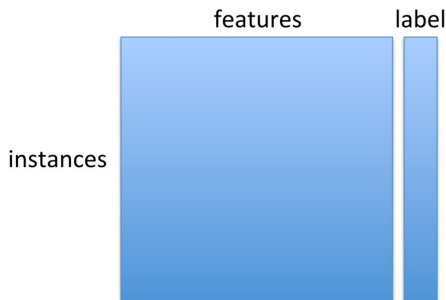
The key challenge of ensemble learning is to obtain models that are:

- reasonably accurate
- as independent as possible

How to achieve independence of models?

Idea 1: Split + Train

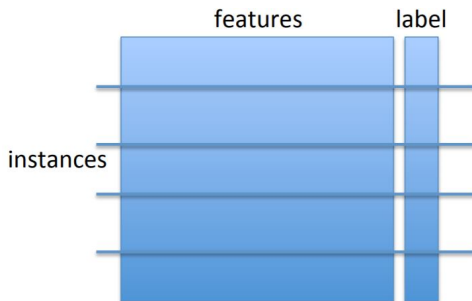
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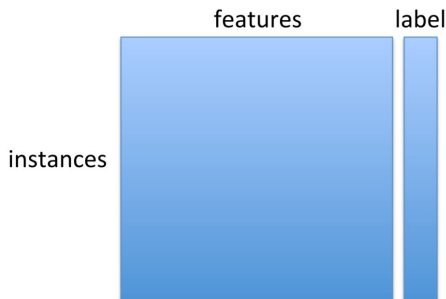
- Randomly split training data into M disjoint groups of instances, train a separate model on each group
- This is not a good idea, since groups get small and the learned models will have poor prediction quality



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Idea 2: Split by features + Train

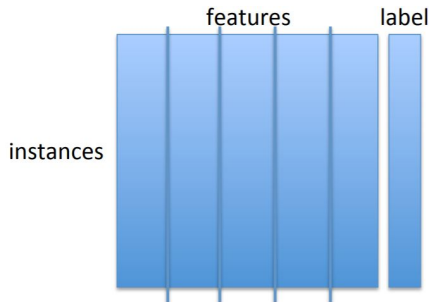
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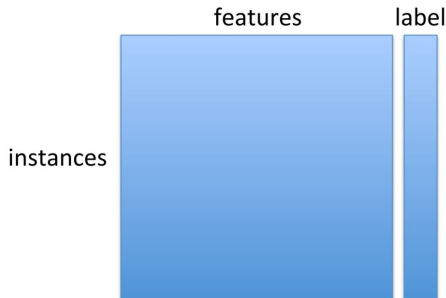
- Randomly split training data into M disjoint groups of features, train a separate model on each group
- This is not a good idea, since not having good features can be even worse than not having enough instances to train



How to achieve independence of models?

Idea 3: Overlapping subsets of instances + Train

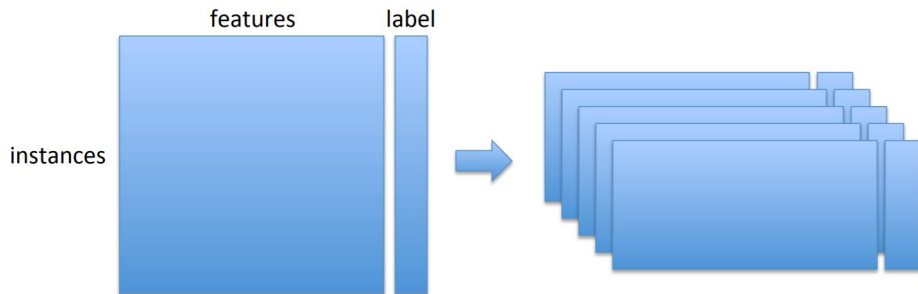
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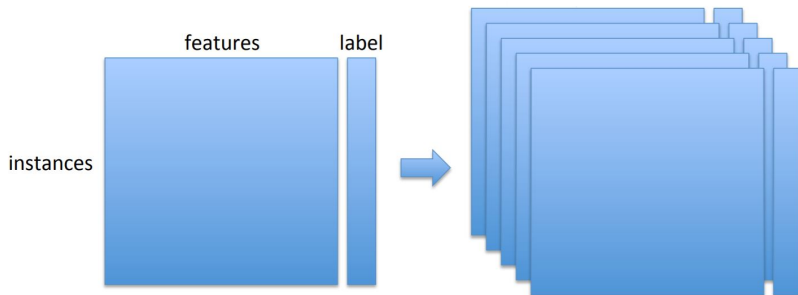
- Randomly sample M overlapping groups of instances, train a separate model on each group
- Not too bad, but still smaller training sets



How to achieve independence of models?

Idea 4: Bootstrap-sampled instances + Train

- Sample with replacement M overlapping groups of instances with same size as original dataset (bootstrapping), train a separate model on each group



Generating a new dataset by Bootstrapping

Bootstrapping:

Sample n items with replacement from the original n training instances

Original data

id	x1	x2	x3	y
1	0.18	0.45	0.8	0
2	0.11	0.82	0.07	0
3	0.87	0.3	0.21	1
4	0.34	0.49	0.18	1
5	0.95	0.64	0.63	0
6	0.03	0.59	0.15	1

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id	x1	x2	x3	y
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Is it possible that a particular training instance is not included in the bootstrap sample?

It is possible with less than 50% probability

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- Bagging (Bootstrap AGGgregating = BAGG):
Train a model separately on each bootstrapped dataset and then aggregate the results.

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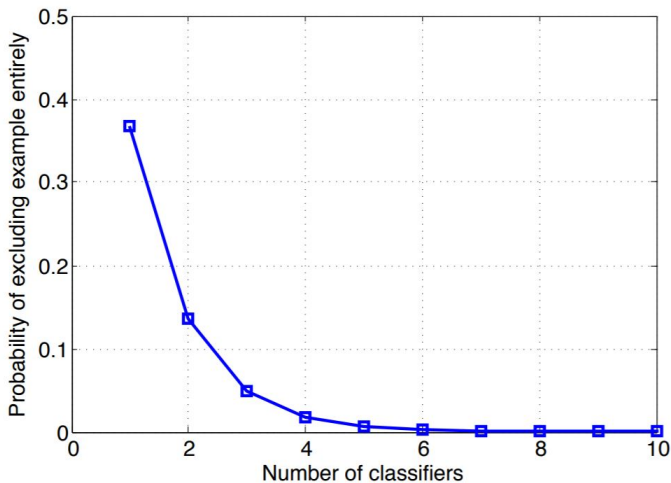
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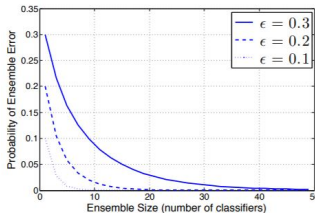
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 - If the probability will tend to 0 as M grows
- Bagging
 - Train a weak classifier on each bootstrap sample and then aggregate the results
 - **It is possible, but happens less and less frequently as M grows**

$\mathbb{P}(\text{excluding an instance from the whole ensemble})$

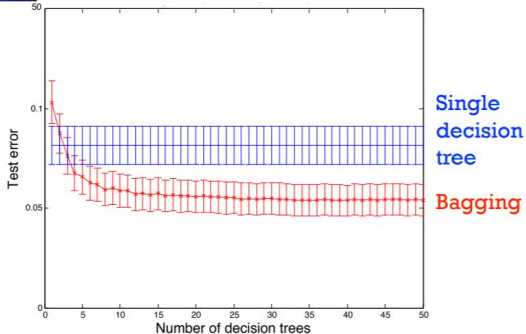


Practical Considerations



← Dream!

Reality! →



Algorithm $\text{Bagging}(D, T, \mathcal{A})$ – train an ensemble of models from bootstrap samples.

Input : data set D ; ensemble size T ; learning algorithm $\mathcal{A}$.

Output : ensemble of models whose predictions are to be combined by voting or averaging.

```
1 for  $t = 1$  to  $T$  do
2   |   build a bootstrap sample  $D_t$  from  $D$  by sampling  $|D|$  data points with
   |   replacement;
3   |   run  $\mathcal{A}$  on  $D_t$  to produce a model  $M_t$ ;
4 end
5 return  $\{M_t | 1 \leq t \leq T\}$ 
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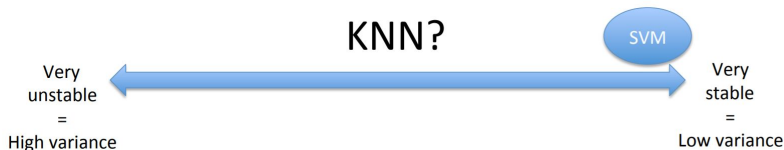
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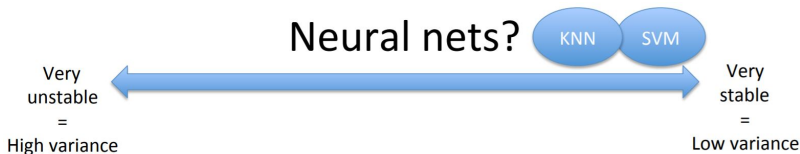
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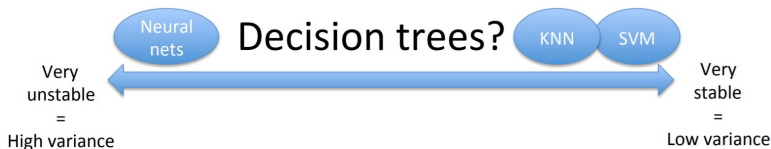
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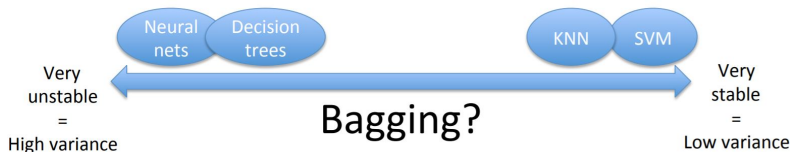
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- Individual models trained on bootstrap-sampled instances, predictions are aggregated
- Bagging is useful when the algorithm to learn individual models is:
 - Relatively accurate
 - Relatively unstable (high variance)
- The aggregated model is then usually better than the original model trained on full dataset

Random Forest (Breiman 2000)

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- **In each recursive step** of learning decision trees:
 - **Randomly select F features** out of all P given features
 - Find the best split among these features
- Parameter F is usually fixed to be $F = \sqrt{P}$ for classification

Random Forest

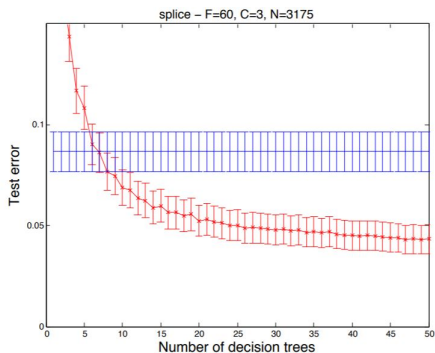
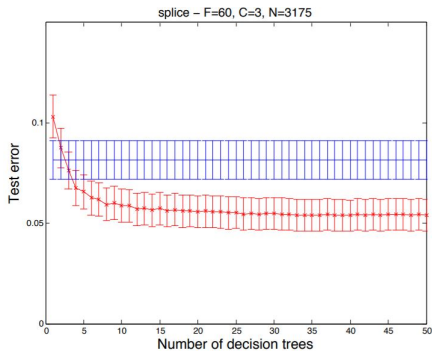
Algorithm $\text{RandomForest}(D, T, d)$ – train an ensemble of tree models from bootstrap samples and random subspaces.

Input : data set D ; ensemble size T ; subspace dimension d .

Output : ensemble of tree models whose predictions are to be combined by voting or averaging.

```
1 for  $t = 1$  to  $T$  do
2   | build a bootstrap sample  $D_t$  from  $D$  by sampling  $|D|$  data points with
   | replacement;
3   | select  $d$  features at random and reduce dimensionality of  $D_t$  accordingly;
4   | train a tree model  $M_t$  on  $D_t$  without pruning;
5 end
6 return  $\{M_t | 1 \leq t \leq T\}$ 
```

Bagging vs Random Forest



Bagging (LEFT) vs Random Forests (RIGHT) on the Splice dataset.

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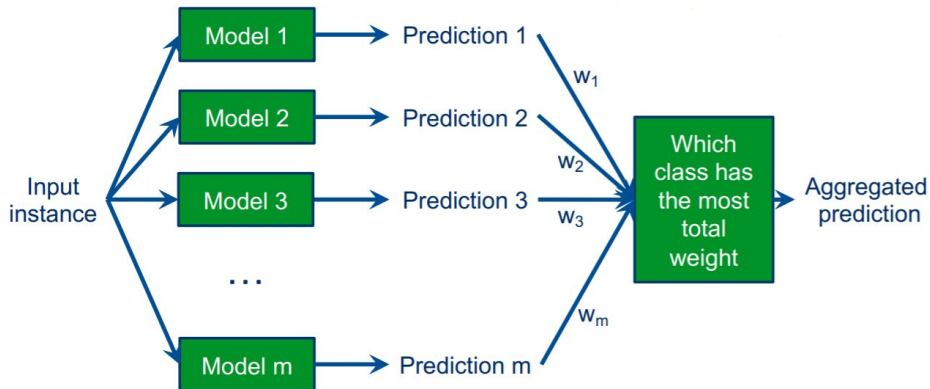
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- Out-of-bag (OOB) error
 - For each training instance make a prediction using trees that do not use that instance and evaluate
 - Stabilisation of OOB error suggests that there are enough trees

Weighted voting



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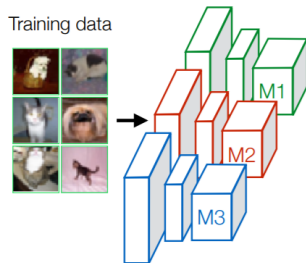
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- How to obtain weights?

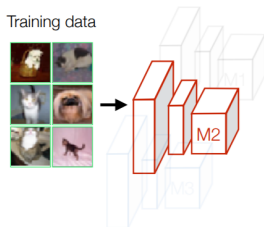
Obtaining weights

We can estimate the weight of each model based on **CV** on training data



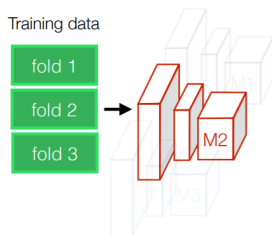
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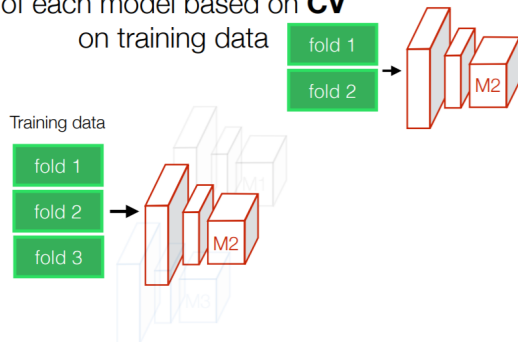
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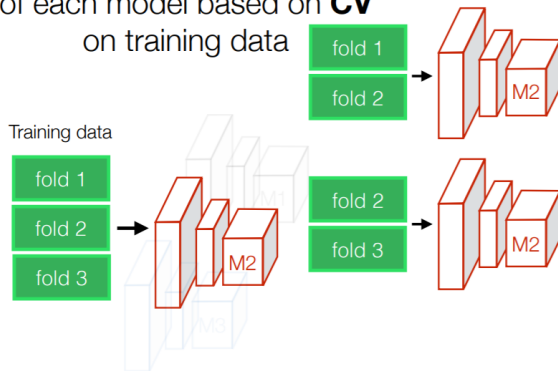
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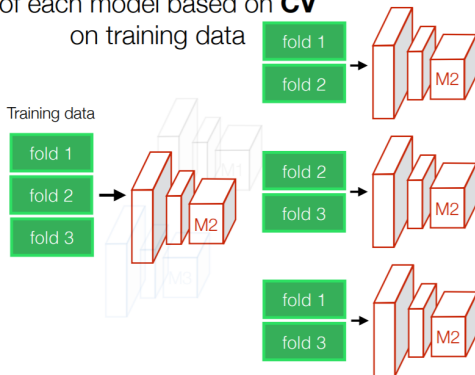
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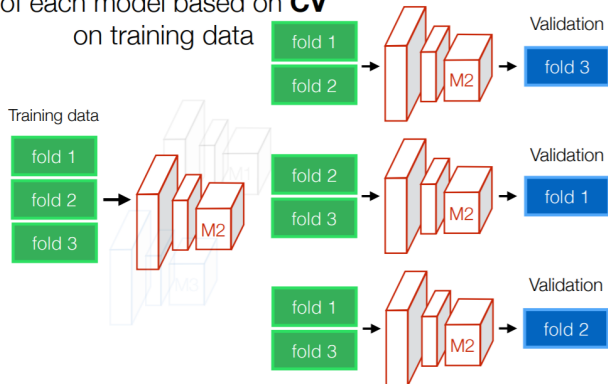
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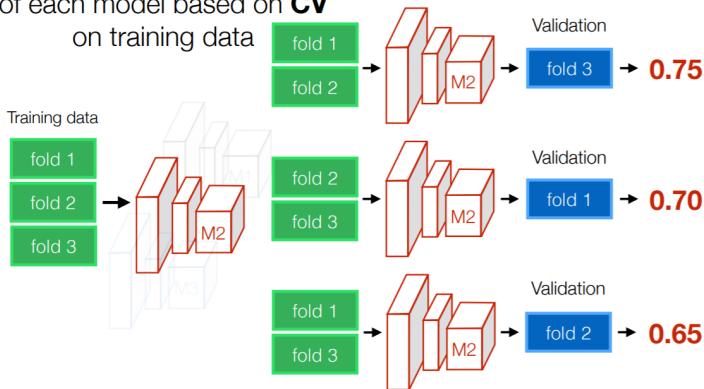
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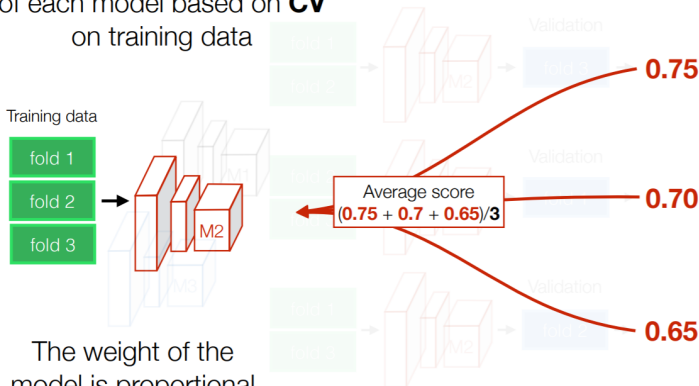
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Obtaining weights

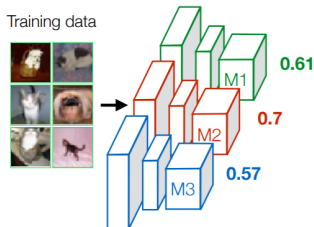
We can estimate the weight of each model based on **CV** on training data



The weight of the model is proportional to its **average score**

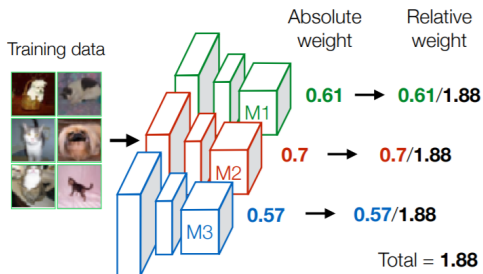
Obtaining weights

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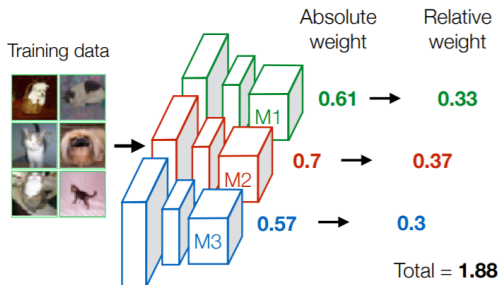
Obtaining weights

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Obtaining weights

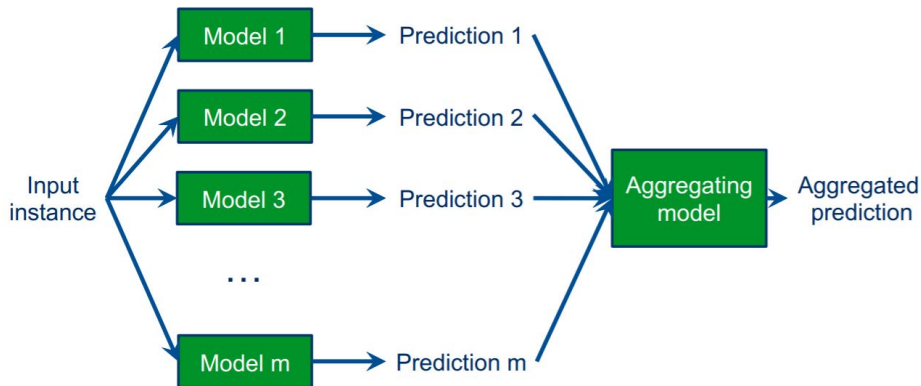
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- We can learn the weights in a linear classification task, where the individual model outputs are treated as features

- We can learn the weights in a linear classification task, where the individual model outputs are treated as features
- This is known as **stacking**, because we stack one classifier on top of many individual classifiers

Stacking



Stacking

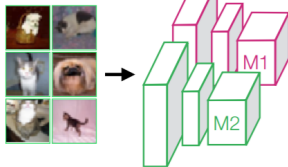


We can train **another model** to infer these weights for **each input point separately**

Stacking

Let's assume we have two models (**M1** and **M2**)

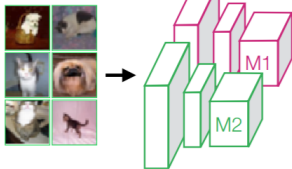
Training data



Stacking

We use models **M1** and **M2** to predict probability of class **dog** for each image

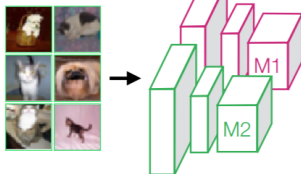
Training data



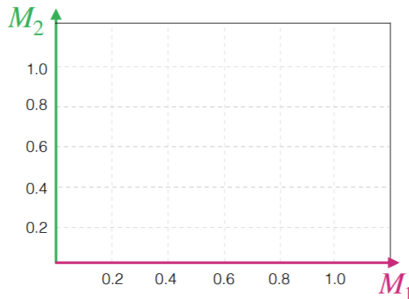
Stacking

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Training data



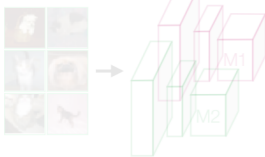
The resulting probabilities will be plotted on the graph below



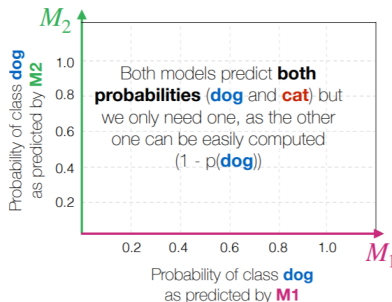
Stacking

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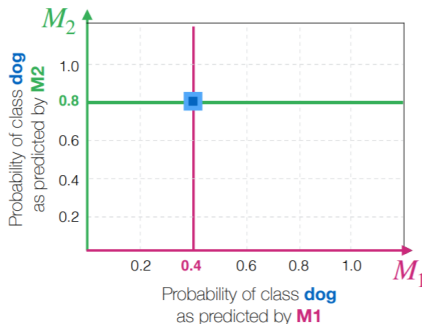
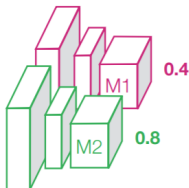
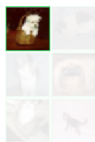


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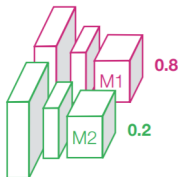
Training data



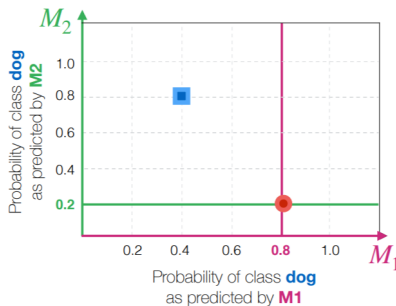
Stacking

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Training data



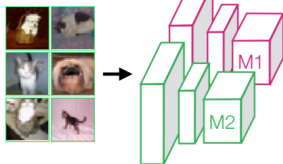
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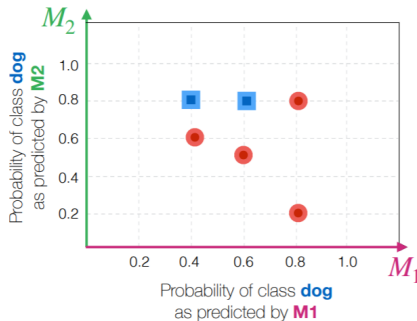
Stacking

We use models **M1** and **M2** to predict probability of class **dog** for each image

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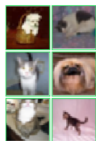
We can train a **Decision Tree** classifier on top of these results



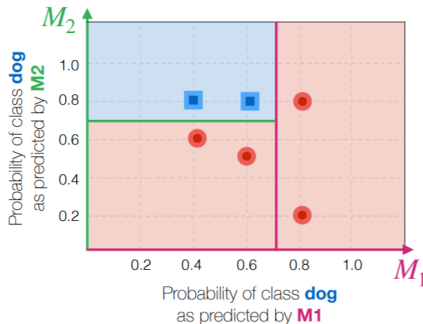
Stacking

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Training data



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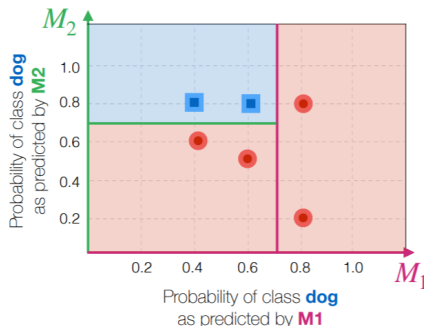
Stacking

We use models **M1** and **M2** to predict probability of class **dog** for each image

Training data



Let's see if we can interpret resulting tree (**meta-model**)

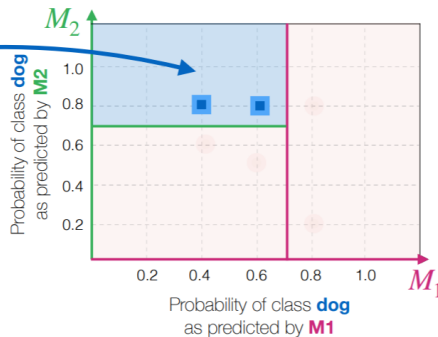


Stacking

We use models **M1** and **M2** to predict probability of class **dog** for each image

Let's see if we can interpret resulting tree (**meta-model**)

If **M2** is confident that image is **dog** ($p > 0.7$) go with its choice

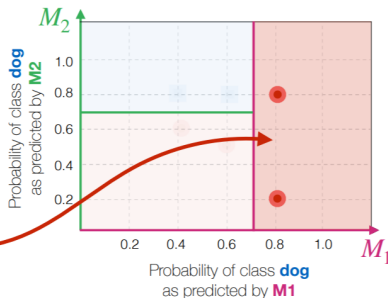


Stacking

We use models **M1** and **M2** to predict probability of class **dog** for each image

Let's see if we can interpret resulting tree (**meta-model**)

If **M2** is confident that image is **dog** ($p > 0.7$) go with its choice
Unless the **M1** is also confident that this image is **dog** ($p > 0.7$)



Stacking

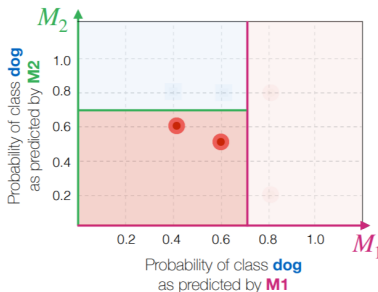
We use models **M1** and **M2** to predict probability of class **dog** for each image

If **M2** is confident that image is **dog** ($p > 0.7$) go with its choice

Unless the **M1** is also confident that this image is **dog** ($p > 0.7$)

If both M1 and M2 are unconfident ($p < 0.7$), predict **cat**

Let's see if we can interpret resulting tree (**meta-model**)

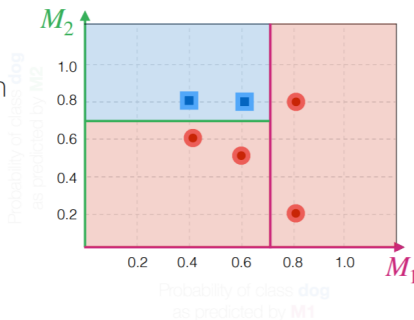


Stacking

Training an additional model (**meta-model**) on top of ensemble predictions is called **stacking**

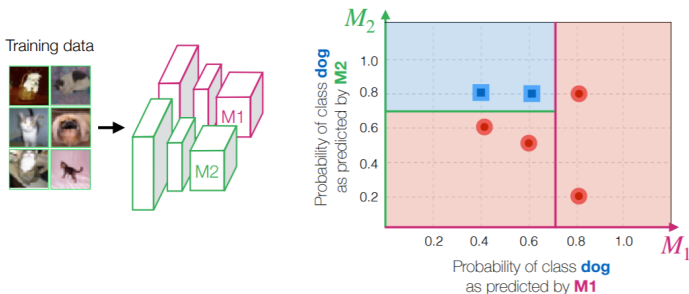
Training data

The **stacking model** can be any model that you find most appropriate for your case



Stacking

Training an additional model (**meta-model**) on top of ensemble predictions is called **stacking**



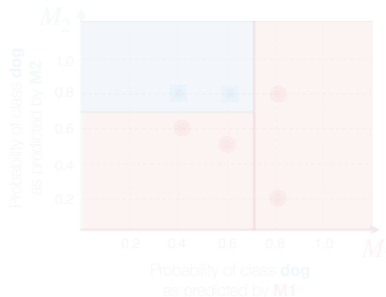
There is **one problem** with the setup I have shown you here

Stacking

Probability scores obtained on training data will be **overfitted**

Training an additional model (**meta-model**) on top of the probability scores obtained on training data is called **stacking**

Training data



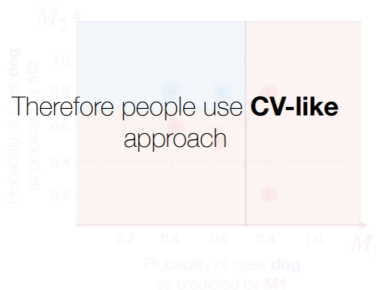
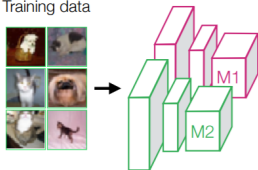
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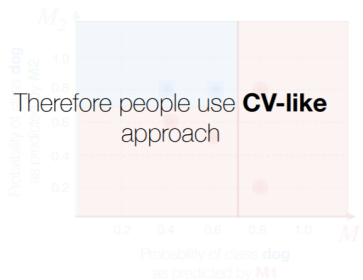
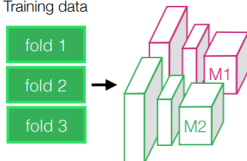


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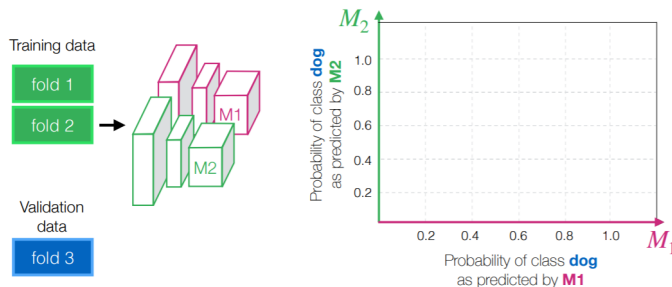
Training an additional model (**meta-model**) on top of
Data is randomly split into folds is called **stacking**

Training data



Stacking

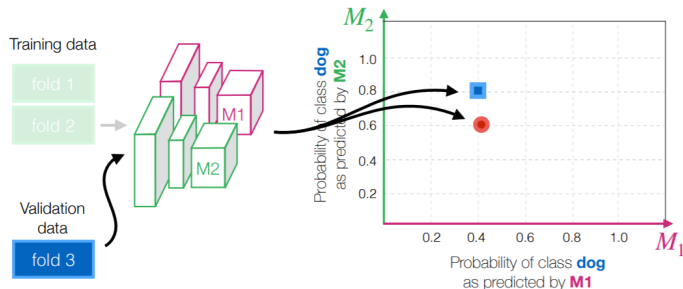
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While models are trained on training data, the probability scores are generated based on validation set

Stacking

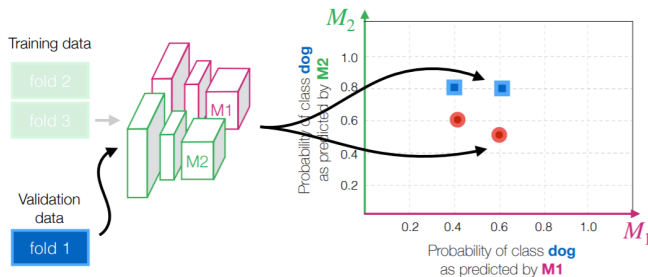
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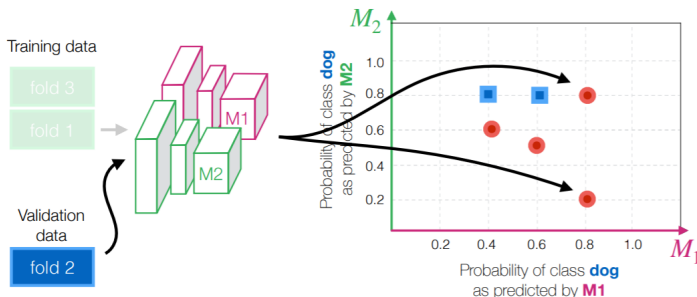
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What have we learned today?

- ✓ Bagging
- ✓ Random Forest
- ✓ Weighted Voting
- ✓ Stacking